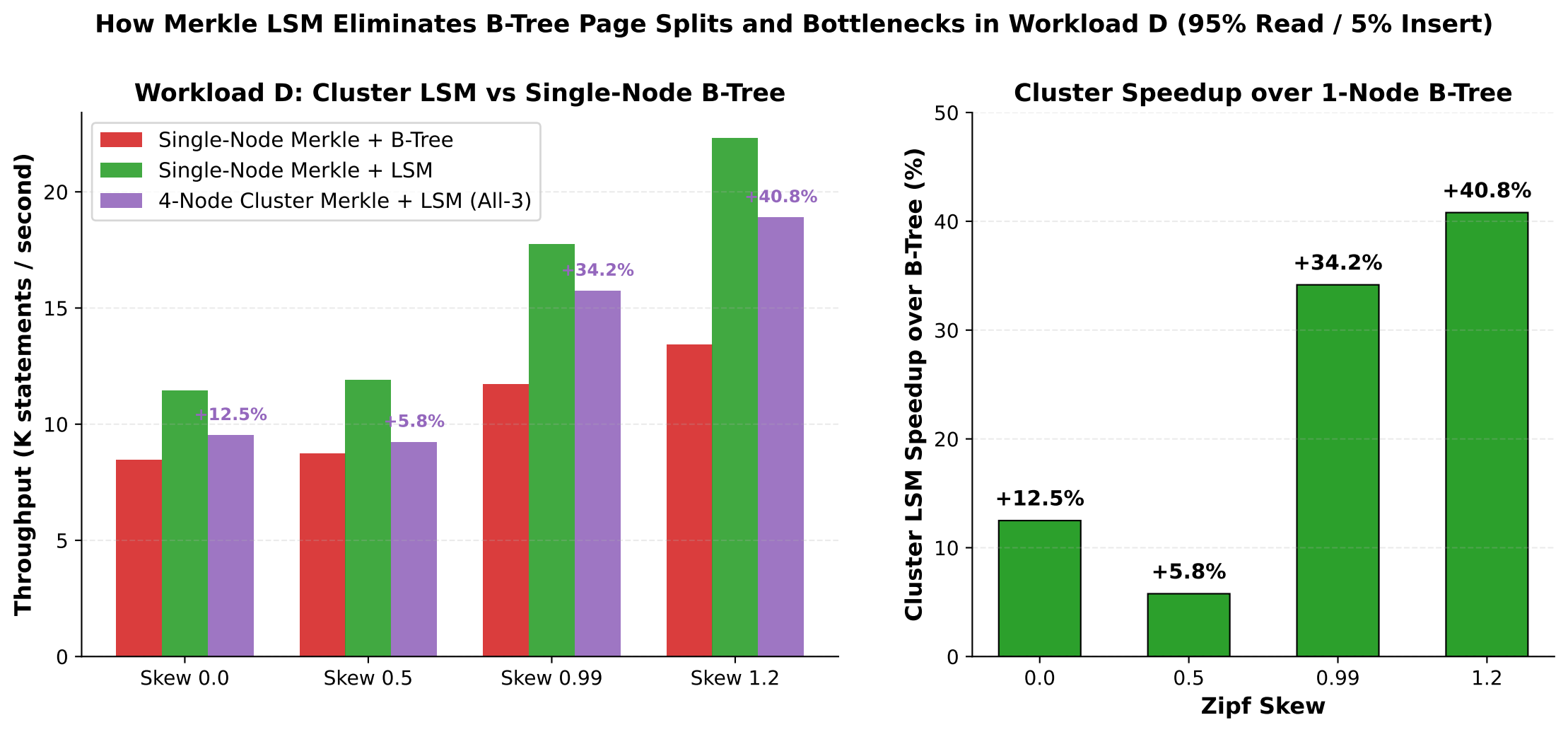
How Merkle LSM Eliminates B-Tree Page Splits and Bottlenecks in Workload D (95% Read / 5% Insert)
Workload D: Cluster LSM vs Single-Node B-Tree
Single-Node Merkle + B-Tree
Single-Node Merkle + LSM
4-Node Cluster Merkle + LSM (All-3)
Throughput (K statements / second)
0
5
10
15
20
+12.5%
+5.8%
+34.2%
+40.8%
Skew 0.0
Skew 0.5
Skew 0.99
Skew 1.2
Cluster Speedup over 1-Node B-Tree
Cluster LSM Speedup over B-Tree (%)
0
10
20
30
40
50
+12.5%
+5.8%
+34.2%
+40.8%
0.0
0.5
0.99
1.2
Zipf Skew